

Target–Coupled CRT Projectors and the Identity Bucket: Shifted Jacobi Grams, Alias Coherence, and the Equality–Recovery Obstruction

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July 2026

Abstract

TPC–37 proposed a target–coupled shifted–fiber estimate as one possible continuation of an orbit–sliced prime–Möbius energy program. We resolve the coefficient–free part of that proposal exactly and identify an additional equality–recovery step required by its physical application.

Let q_1, q_2, q_3 be distinct orbit–scale primes, $M = q_1 q_2 q_3$, $\Phi = \varphi(M)$, and let N be a unit modulo M . If $F = su - N$, every coordinate face has the exact expansion

$$K_{S,N}(F) = \frac{1}{\Phi} \sum_{\chi \in \mathfrak{X}_S} \chi(s) \chi(u) \overline{\chi(N)},$$

where the local component of χ is nonprincipal exactly on S . Zero extension makes this expansion valid even when $N + F$ is a nonunit; then every face vanishes in a killed local coordinate. On the regular support $N + F \in (\mathbb{Z}/M\mathbb{Z})^\times$, the finite CRT target dependence is diagonalized by the unit–group ratio

$$r = (N + F)N^{-1} \pmod{M}.$$

The full and proper face families are complementary character projectors. Their recombination is the target–independent identity

$$C_N(F) + P_N(F) = \mathbf{1}_{M|F}.$$

On this regular support we prove a Hilbert–valued ratio Parseval identity. For arbitrary vectors $z_{N,F}$, character orthogonality controls the sums of the coherent ratio buckets, not the individual pairs (N, F) . For fixed F , the ratio map has exact maximum multiplicity

$$d_M(F) = \prod_{q_i | F} (q_i - 1).$$

When $q_i \asymp J$, the target collision scales are therefore $1, J, J^2, J^3$, according as F is divisible by zero, one, two, or three auxiliary primes. Generic determinants are injective; the obstruction is concentrated on the coordinate divisibility strata.

We also compute the complete one–period target–synthesis spectrum. The full–face operator has squared norm

$$\prod_{i=1}^3 \frac{q_i(q_i - 2)^2}{(q_i - 1)^2} \asymp \Phi,$$

whereas, for $q_i \asymp J$, the proper operator has squared norm $\asymp \Phi/J$. Both scalar spectra have a dominant global principal target mode; fixed-target Parseval does not imply target-synthesis orthogonality.

After full/proper recombination, every nonjoint ghost cancels exactly and the completed output is the joint-alias sum over $F = kM$. At the endpoint

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad M = X^{399/400+o(1)},$$

their endpoint order is

$$\mathcal{K}_X := 1 + X/M = X^{1/400+o(1)}.$$

All multiplicative character phases equal 1 on this identity bucket. For the alias-completed sum, diagonal energy plus Cauchy loses the exact finite alias count; at the endpoint this has order $\mathcal{K}_X$. A joint-alias Gram estimate removes that loss.

Crucially, the alias-completed sum is not the original equality output, which is its $k = 0$ column. There is no coefficient-free recovery: two opposite alias columns make both fixed full/proper field outputs vanish while leaving the equality column arbitrary. Thus the shifted-fiber estimate proposed in TPC-37 controls a congruence completion but, by itself, does not imply the physical equality gate. A full alias-Gram bound implies the equality bound directly through its $k = 0$ diagonal entry, at the stronger diagonal scale. The terminal identity bucket expands into Möbius autocorrelations at shifts rM , and the physical row expansion contains the affine four-Möbius boundary found earlier.

Finally, we distinguish two prescribed face-field estimates from an all-field operator estimate of the same normalization that is stable under restrictions. The latter would be strictly stronger: its $F = 0, s = 1$ restriction implies a diagonal-scale estimate for the terminal affine-Möbius/four-Möbius operator. We prove no positive physical equality-recovery estimate and no Möbius estimate. The results are exact reductions, a correction of the logical gate, and method obstructions—not a Möbius lower bound, a parity breach, a Hardy-Littlewood asymptotic, or a twin-prime theorem.

Keywords: Möbius correlation; Dirichlet character; CRT projector; Jacobi sum; Bessel sequence; alias Gram; equality recovery; parity boundary.

MSC 2020: Primary 11N35, 11L40; Secondary 11N37, 15A18, 42C15.

1 Introduction

The orbit-sliced program in TPC-34–TPC-37 reduces a scalar prime-pair remainder to a coefficient-specific energy estimate for a reflected Möbius divisor sum [4–7]. The physical row mask is now known to have subpolynomial signed projective complexity, the zero residue faces have been removed at the target scale, and all seven proper faces of a three-modulus completion close on the exact equality. The full face remains unsuppressed there.

TPC-37 recorded two possible continuations. The first works after finite-field completion and asks for an actual weighted Mellin-overlap estimate, including its principal boundary. The second keeps the integer determinant

$$F = su - N \tag{1}$$

visible and asks for a target-coupled shifted-fiber estimate. This paper audits the second continuation. The question is not whether one may write another formal Hilbert-space norm. It is whether the moving target and the CRT faces create genuine orthogonality at the physical normalization.

1.1 What the target coupling really is

Let q_1, q_2, q_3 be distinct odd primes, put

$$M = q_1 q_2 q_3, \quad \Phi = \varphi(M) = \prod_{i=1}^3 (q_i - 1), \quad (2)$$

and let N be a unit modulo M . TPC-37 defines, in each local coordinate, a principal face $m_{i,N}(F)$ and a nonprincipal aggregate $c_{i,N}(F)$. Their tensor products give eight faces. The full face C_N uses three nonprincipal coordinates; the proper union P_N contains the other seven.

The first result of this paper is the exact identity

$$K_{S,N}(F) = \frac{1}{\Phi} \sum_{\chi \in \mathfrak{X}_S} \chi(N+F) \overline{\chi(N)}, \quad (3)$$

where the local component of χ is nonprincipal exactly for $i \in S$. Since $N+F = su$, the phase factors further as

$$\chi(N+F) \overline{\chi(N)} = \chi(s) \chi(u) \overline{\chi(N)}. \quad (4)$$

Zero-extended characters make this valid without a hidden correction when s or u is nonunit; then every face vanishes in a killed local coordinate. On the regular support $N+F \in (\mathbb{Z}/M\mathbb{Z})^\times$, the finite CRT target dependence is diagonalized exactly by the unit-group ratio

$$r = \frac{N+F}{N} \pmod{M}. \quad (5)$$

For fixed F , on its regular domain this ratio map is injective in every local coordinate in which $q_i \nmid F$, and is constant in every coordinate in which $q_i \mid F$. Its exact largest fiber is

$$d_M(F) = \prod_{q_i \mid F} (q_i - 1). \quad (6)$$

When $q_i \asymp J$, this gives the target collision scale ladder

$$1, \quad J, \quad J^2, \quad J^3, \quad (7)$$

according as F is divisible by zero, one, two, or three orbit-scale primes. The underlying multiplicity formula is exact; the displayed J -powers are asymptotic scales, not a large-sieve heuristic.

1.2 Fixed target orthogonality does not survive target synthesis

For one fixed N , the functions

$$e_{\chi,N}(F) = \chi(N+F) \overline{\chi(N)} \quad (8)$$

are orthogonal over $F \bmod M$. This recovers the one-period masses

$$\sum_{F \bmod M} |C_N(F)|^2 = A, \quad \sum_{F \bmod M} |P_N(F)|^2 = 1 - A, \quad (9)$$

where

$$A = \prod_{i=1}^3 \left(1 - \frac{1}{q_i - 1}\right). \quad (10)$$

When N moves, the cross Gram becomes a product of shifted Jacobi sums. More directly, the full face rows satisfy

$$\langle C_N, C_{N'} \rangle = \prod_{i=1}^3 \begin{cases} 1 - p_i, & N \equiv N' \pmod{q_i}, \\ 1 - p_i - p_i^2, & N \not\equiv N' \pmod{q_i}, \end{cases} \quad (11)$$

with $p_i = (q_i - 1)^{-1}$. Distinct target rows therefore have inner product $1 - O(J^{-1})$ when they are distinct in all coordinates. They are almost parallel, not almost orthogonal.

We compute the entire scalar target-synthesis spectrum. For

$$(T_C a)(F) = \sum_{N \bmod M}^* a_N C_N(F), \quad (12)$$

the squared singular values are indexed by target Mellin modes. The largest is

$$\|T_C\|_{\text{op}}^2 = \prod_{i=1}^3 \frac{q_i(q_i - 2)^2}{(q_i - 1)^2} \asymp \Phi \asymp J^3. \quad (13)$$

For the proper union, the principal squared singular value is $\asymp \Phi/J \asymp J^2$; all its nonprincipal target modes are $O(1)$. The small fixed-row norm $1 - A \asymp J^{-1}$ therefore does not imply a small synthesis norm.

1.3 Recombination and the identity bucket

The full and proper character families are complementary. Summing them before estimating gives

$$\boxed{C_N(F) + P_N(F) = \mathbf{1}_{M|F}}. \quad (14)$$

Every nonjoint ghost cancels. Separate estimates for C_N and P_N are consequently sufficient but strictly stronger than an estimate for their recombined congruence completion.

In the physical determinant interval, (14) leaves only

$$F = kM, \quad |k| \ll X/M. \quad (15)$$

At the endpoint

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad M = X^{399/400+o(1)}, \quad (16)$$

the endpoint alias order is

$$\mathcal{K}_X := 1 + \frac{X}{M} = \frac{Q}{J^2} X^{o(1)} = X^{1/400+o(1)}. \quad (17)$$

On $F = kM$, the ratio in (5) is 1 and every character phase in (3) equals one. Multiplicative character orthogonality is therefore exactly blind on the remaining bucket.

Write $\mathcal{Z}_k$ for the raw shifted output on $su = N + kM$. The recombined completed output is

$$\left\| \sum_k \mathcal{Z}_k \right\|_2^2 = \sum_{k,k'} \langle \mathcal{Z}_k, \mathcal{Z}_{k'} \rangle. \quad (18)$$

If its alias columns satisfy the diagonal trace bound

$$\sum_k \|\mathcal{Z}_k\|_2^2 \ll X^{o(1)} Q^2 J \mathcal{K}_X, \quad (19)$$

Cauchy gives only $X^{o(1)}Q^2JK_X^2$. The desired scale is

$$Q^2JK_X = \frac{Q^3}{J}X^{o(1)}. \quad (20)$$

The exact finite loss is the alias count K_B ; at the endpoint it is of order K_X , and a rank-one family of alias columns attains it.

There is a more basic logical distinction. The original physical output is $\mathcal{Z}_0$, not the alias sum in (18). No norm bound for the latter recovers the former: for two aliases, the abstract choice $\mathcal{Z}_0 = v$, $\mathcal{Z}_1 = -v$ makes the alias sum and both fixed full/proper field outputs vanish while $\mathcal{Z}_0$ is arbitrary. Thus the shifted-fiber estimate proposed in TPC-37 controls the congruence-completed output but needs an additional equality-recovery input before it can imply the original orbit gate. A uniform column bound at scale Q^2J would recover $\mathcal{Z}_0$, but at $k = 0$ it is already stronger than the desired Q^2JK_X bound. A full alias-Gram bound is stronger still and automatically contains this column bound.

1.4 Arithmetic meaning and claim boundary

On the terminal submodel $s = 1$, a joint alias contains

$$a_N(k) = -\mu(N + kM) \log(N + kM) \Phi_0(N + kM). \quad (21)$$

Its squared alias sum has cross terms

$$\mu(N + kM)\mu(N + (k + r)M), \quad (22)$$

and restoring the physical rows gives the affine four-Möbius Gram identified in TPC-34. This is an averaged, highly structured correlation. It is not logically equivalent to pointwise Chowla in each short progression.

There is also a necessary distinction in terminology. Bounds for the two prescribed fields C and P do not constitute an all-field operator estimate. An estimate of the same normalization for every coefficient field, stable under fiber restriction, is strictly stronger. Since $C_N(0) = A = 1 - O(J^{-1})$, that hereditary all-field estimate restricted to $F = 0$, $s = 1$ would imply a diagonal-scale estimate for the original terminal operator, stronger than the inherited target by the factor K_X . We prove no hereditary all-field theorem, positive physical equality-recovery estimate, or Möbius correlation estimate.

The unconditional conclusions of the paper are the exact character projectors, shifted Jacobi Grams, ratio Parseval theorem, exact collision multiplicity and its scale ladder, target-synthesis spectrum, face recombination, equality-nonrecoverability, and sharp abstract coherence examples. They locate the remaining gates but do not prove the physical orbit estimate, affine Chowla, sieve-parity failure, a Hardy-Littlewood asymptotic, or infinitely many twin primes.

Section 2 fixes the inherited physical normalization. Section 3 proves the target-coupled character calculus. Section 4 gives the ratio and determinant stratifications. Section 5 computes the synthesis spectrum. Section 6 reduces to the identity bucket. Section 7 records its arithmetic content, and sections 8 and 9 state the next inputs and formal claim boundary.

2 Inherited physical interface and normalizations

The finite CRT statements later in the paper are unconditional. Their application to the orbit-sliced prime-Möbius packet uses only the interface below, inherited from TPC-34-TPC-37 [4-7].

2.1 Rows, targets, and the orbit gate

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and work on one retained dyadic packet with

$$Q = LD, \quad QJ \asymp X, \quad R \leq T < U_0 \asymp X. \quad (23)$$

A physical row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad (24)$$

where $\ell_\alpha \asymp L$ is prime and $d_\alpha \asymp D$ is squarefree. Source separation makes m_α determine the row. There are $O(QX^{o(1)})$ rows. The two physical coefficient systems satisfy

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll X^{o(1)} Q, \quad (25)$$

$$\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll X^{o(1)} Q, \quad i = 1, 2. \quad (26)$$

They contain the actual squarefree Möbius and prime von-Mangoldt provenance; they are not arbitrary bounded row vectors.

For j in a positive interval $\mathcal{J}$ of $O(J)$ integers, put

$$N_\alpha(j) = m_\alpha j + h \asymp X. \quad (27)$$

Choose the distinct auxiliary primes $q_i \asymp J$ under the inherited zero-face hypotheses

$$\max \mathcal{J} < q_i, \quad \max_\alpha d_\alpha < q_i < \min_\alpha \ell_\alpha, \quad q_i \nmid h, \quad (28)$$

and put $M = \prod_i q_i$. Let $\mathcal{N}_{\text{reg}}$ be the subpacket on which

$$(N_\alpha(j), M) = 1. \quad (29)$$

The zero-face estimates of TPC-37 place the complementary target rows at the inherited acceptable scale. From this point onward, all shifted outputs and the symbols V_L, V_R refer to this regular subpacket. Thus the character and ratio calculations below are applied only where their unit hypotheses hold; adjoining the already controlled complement recovers the full packet ledger.

The ultra increment is

$$C_m(j) = \sum_{\substack{T < u \leq U_0 \\ u \mid mj+h}} -\mu(u) \log u. \quad (30)$$

On the physical support,

$$(u, mjH) = 1 \quad (31)$$

for every divisor in this sum.

The actual retained multiplier has the form

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) = \mathfrak{m}_{\text{act}}(\alpha, \gamma) \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J), \quad (32)$$

where the static mask is

$$\mathfrak{m}_{\text{act}}(\alpha, \gamma) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}. \quad (33)$$

TPC-36 proves an $X^{o(1)}$ signed projective reassembly for this mask and the fixed residue and smooth orbit factors. It does not prove an arithmetic Gram estimate for the resulting divisor sums.

Define

$$Y_{\gamma,j}^L = \sum_{\alpha} \gamma_{\alpha}^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\alpha}}(j), \quad V_L = \sum_{\gamma,j} |Y_{\gamma,j}^L|^2, \quad (34)$$

$$Y_{\alpha,j}^R = \sum_{\gamma} \gamma_{\gamma}^{(2)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\gamma}}(j), \quad V_R = \sum_{\alpha,j} |Y_{\alpha,j}^R|^2. \quad (35)$$

The inherited scalar remainder closes once

$$\boxed{V_L + V_R \ll X^{o(1)} \frac{Q^3}{J}}. \quad (36)$$

The proved input–copy diagonal is

$$V_{L,\Delta} + V_{R,\Delta} \ll X^{o(1)} Q^2 J. \quad (37)$$

This diagonal is attached to the original fixed shift h .

2.2 Shifted determinant fibers

Opening one ultra divisor and its complementary factor gives

$$su = N_{\alpha}(j) = \ell_{\alpha} d_{\alpha} j + h. \quad (38)$$

After dyadic and projective separation, a layer has the schematic integer form

$$\sum_{\ell,d,u,s} b(\ell) f(d) g(u) w(s) \mathbf{1}_{su=\ell jd+h}, \quad (39)$$

with the physical ultra coefficient

$$g(u) = -\mu(u) \log u \Phi_U(u). \quad (40)$$

For a shifted determinant F , replace (38) by

$$su = N_{\alpha}(j) + F = \ell_{\alpha} d_{\alpha} j + h + F. \quad (41)$$

Let

$$\mathcal{H}_{\text{phys}} = \ell^2\{(L, \gamma, j)\} \oplus \ell^2\{(R, \alpha, j)\} \quad (42)$$

be the retained left/right output space. We write $\mathcal{Z}_F[\kappa]$ for the Hilbert–valued shifted output in which an atom with target N is multiplied by $\kappa_N(F)$ before the row sum. This notation contains the already proved $X^{o(1)}$ projective reassembly cost. Write

$$\mathcal{R}_F := \mathcal{Z}_F[\mathbf{1}] \quad (43)$$

for the raw shifted column. The original regular–subpacket output is $\mathcal{R}_0$, and

$$\|\mathcal{R}_0\|_2^2 = V_L + V_R \quad (44)$$

at the packet level, with the declared reassembly convention.

The TPC–37 continuation was the pair of fixed–field estimates

$$\left\| \sum_{|F| \ll X} \mathcal{Z}_F[\kappa] \right\|_2^2 \ll X^{o(1)} Q^2 J \sup_N \sum_{|F| \ll X} |\kappa_N(F)|^2, \quad \kappa = C, P. \quad (45)$$

It was proposed there as a sufficient conditional input. The present paper does not assume it and refines its logical scope: the displayed estimate controls a face–weighted congruence completion. Recovering $\mathcal{R}_0$ from that completion requires an additional statement. We determine the exact coefficient–free information available for its two fields, prove the recovery obstruction, and distinguish (45) from an operator inequality valid for arbitrary κ .

2.3 Endpoint ledger

At the endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}, \quad (46)$$

so $QJ = X^{1+o(1)}$. Three pairwise coprime orbit–scale primes give

$$M = q_1 q_2 q_3 = X^{399/400+o(1)}, \quad \mathcal{K}_X := 1 + \frac{X}{M} = X^{1/400+o(1)}. \quad (47)$$

The key identity is

$$Q^2 J \mathcal{K}_X = Q^2 J \frac{Q}{J^2} X^{o(1)} = X^{o(1)} \frac{Q^3}{J}. \quad (48)$$

Four orbit–scale primes have product $J^4 > X^{1+o(1)}$ and eliminate nonzero joint aliases. As shown later, they do not weaken the full face on the equality $F = 0$.

3 Target–coupled CRT faces as character projectors

We first work with any fixed number $r \geq 1$ of distinct odd primes. Put

$$M = \prod_{i=1}^r q_i, \quad n_i = q_i - 1, \quad p_i = n_i^{-1}, \quad \Phi = \varphi(M) = \prod_{i=1}^r n_i. \quad (49)$$

Every Dirichlet character modulo M is viewed as a tensor product of local multiplicative characters and is extended by zero away from the unit group. Throughout this section, $N \in (\mathbb{Z}/M\mathbb{Z})^\times$.

3.1 Coordinate faces

For $F \in \mathbb{Z}/M\mathbb{Z}$, define locally

$$d_{i,N}(F) = \mathbf{1}_{F=0 \pmod{q_i}}, \quad (50)$$

$$e_{i,N}(F) = \mathbf{1}_{N+F \neq 0 \pmod{q_i}}, \quad (51)$$

$$m_{i,N}(F) = p_i e_{i,N}(F), \quad c_{i,N}(F) = d_{i,N}(F) - m_{i,N}(F). \quad (52)$$

The exclusion in (51) is forced by zero extension of multiplicative characters. For $S \subseteq [r]$, put

$$K_{S,N}(F) = \prod_{i \in S} c_{i,N}(F) \prod_{i \notin S} m_{i,N}(F). \quad (53)$$

Let $\widehat{G}_i$ be the characters of $G_i = \mathbb{F}_{q_i}^\times$, including the principal character $\mathbf{1}_i$. Define

$$\mathfrak{X}_S = \left\{ \chi = \bigotimes_{i=1}^r \chi_i : \chi_i \neq \mathbf{1}_i \iff i \in S \right\}. \quad (54)$$

Theorem 3.1 (Exact target–coupled character projector). *For every $S \subseteq [r]$, every unit $N \bmod M$, and every $F \bmod M$,*

$$K_{S,N}(F) = \frac{1}{\Phi} \sum_{\chi \in \mathfrak{X}_S} \chi(N+F) \overline{\chi(N)}. \quad (55)$$

If $N + F = su$, then each summand factors as

$$\chi(N+F) \overline{\chi(N)} = \chi(s) \chi(u) \overline{\chi(N)}. \quad (56)$$

Both identities remain valid when $N + F$, s , or u is a nonunit.

Proof. In one local coordinate, multiplicative orthogonality gives

$$d_{i,N}(F) = \frac{1}{n_i} \sum_{\chi_i \in \widehat{G}_i} \chi_i(N+F) \overline{\chi_i(N)}. \quad (57)$$

If $N+F=0$, both sides vanish. The principal term on the right is $m_{i,N}(F)$, because N is a unit. Subtracting it gives

$$c_{i,N}(F) = \frac{1}{n_i} \sum_{\substack{\chi_i \in \widehat{G}_i \\ \chi_i \neq \mathbf{1}_i}} \chi_i(N+F) \overline{\chi_i(N)}. \quad (58)$$

Tensor (57) and (58) through the CRT to prove (55). Complete multiplicativity gives (56). If one factor is a nonunit, the relevant character values vanish on both sides. $\square$

For $r=3$, define

$$\mathfrak{X}_{\text{full}} = \mathfrak{X}_{[3]}, \quad \mathfrak{X}_{\text{prop}} = \bigcup_{S \subsetneq [3]} \mathfrak{X}_S, \quad (59)$$

and

$$C_N = K_{[3],N}, \quad P_N = \sum_{S \subsetneq [3]} K_{S,N}. \quad (60)$$

With

$$A = \prod_{i=1}^3 (1 - p_i), \quad (61)$$

their family sizes are

$$|\mathfrak{X}_{\text{full}}| = A\Phi, \quad |\mathfrak{X}_{\text{prop}}| = (1-A)\Phi. \quad (62)$$

Corollary 3.2 (Full/proper recombination). *For every unit target N modulo M ,*

$$\boxed{C_N(F) + P_N(F) = \mathbf{1}_{M|F}.} \quad (63)$$

In particular,

$$C_N(0) = A, \quad P_N(0) = 1 - A. \quad (64)$$

If $M \nmid F$, then

$$P_N(F) = -C_N(F). \quad (65)$$

Proof. The two families in (59) partition all characters modulo M . Apply character orthogonality to their sum. At $F=0$, all character phases equal one, so the two values are their family densities. The final identity follows from the first. $\square$

3.2 Fixed target Parseval and shifted target Gram

For a fixed unit N , put

$$e_{\chi,N}(F) = \chi(N+F) \overline{\chi(N)}. \quad (66)$$

Proposition 3.3 (Fixed target Parseval). *For characters $\chi, \psi \bmod M$,*

$$\sum_{F \bmod M} e_{\chi,N}(F) \overline{e_{\psi,N}(F)} = \Phi \mathbf{1}_{\chi=\psi}. \quad (67)$$

Consequently,

$$\boxed{\sum_{F \bmod M} |C_N(F)|^2 = A, \quad \sum_{F \bmod M} |P_N(F)|^2 = 1 - A.} \quad (68)$$

Proof. The translation $F \mapsto N + F$ runs through every residue modulo M . Character orthogonality on the unit group proves (67). Insert the disjoint character families and their sizes from (62). $\square$

When the target moves, the Gram is no longer diagonal. For units N, N' let $\Delta = N' - N$ and define

$$\Gamma_{\chi, \psi}(N, N') = \sum_{F \bmod M} e_{\chi, N}(F) \overline{e_{\psi, N'}(F)}. \quad (69)$$

Theorem 3.4 (Shifted Jacobi Gram). *With local components denoted by subscripts,*

$$\Gamma_{\chi, \psi}(N, N') = \overline{\chi(N)} \overline{\psi(N')} \prod_{i=1}^r S_i(\chi_i, \psi_i; \Delta_i), \quad (70)$$

where

$$S_i(\chi_i, \psi_i; \Delta_i) = \sum_{x \bmod q_i} \chi_i(x) \overline{\psi_i(x + \Delta_i)}. \quad (71)$$

If $\Delta_i = 0$, then

$$S_i = n_i \mathbf{1}_{\chi_i = \psi_i}. \quad (72)$$

If $\Delta_i \neq 0$, then

$$S_i = \chi_i(-\Delta_i) \overline{\psi_i(\Delta_i)} J(\chi_i, \overline{\psi_i}), \quad (73)$$

where $J(\alpha, \beta) = \sum_x \alpha(x) \beta(1 - x)$.

Proof. Set $x = N + F$. Then $N' + F = x + \Delta$, and the CRT factors the sum in (69) into (71). The zero-shift case is ordinary character orthogonality. If $\Delta_i \neq 0$, substitute $x = -\Delta_i y$. Then $x + \Delta_i = \Delta_i(1 - y)$, which gives (73). $\square$

Remark 3.5 (What has and has not been diagonalized). The moving affine determinant has been converted into an exact ratio phase. The price is the shifted Jacobi Gram (70); fixed-target Parseval does not persist when target rows are synthesized coherently. This distinction is the finite spectral content of the target-coupling problem.

4 Hilbert-valued ratio Parseval and determinant strata

The correct large-sieve variable is the ratio, not F or N separately. This section gives the exact Hilbert-valued identity and its sharp collision multiplicities. The general large sieve may be compared with [1, 3]; no asymptotic large-sieve input is needed here.

4.1 The ratio bucket identity

Let $\mathcal{I}$ be a finite multiset of pairs (N, F) such that both N and $N + F$ are units modulo M . Attach vectors $z_{N, F} \in \mathcal{H}$, where $\mathcal{H}$ is a complex Hilbert space, and put

$$r(N, F) = (N + F)N^{-1} \in (\mathbb{Z}/M\mathbb{Z})^\times. \quad (74)$$

For every ratio r , define its coherent bucket

$$V_r = \sum_{\substack{(N, F) \in \mathcal{I} \\ r(N, F) = r}} z_{N, F}. \quad (75)$$

For a character $\chi \bmod M$, put

$$Y_\chi = \sum_{(N, F) \in \mathcal{I}} \chi(N + F) \overline{\chi(N)} z_{N, F}. \quad (76)$$

Theorem 4.1 (Hilbert-valued ratio Parseval). *One has the exact identity*

$$\boxed{\sum_{\chi \bmod M} \|Y_\chi\|_{\mathcal{H}}^2 = \Phi \sum_{r \in (\mathbb{Z}/M\mathbb{Z})^\times} \|V_r\|_{\mathcal{H}}^2.} \quad (77)$$

For any character family $\mathfrak{A}$, define

$$K_{\mathfrak{A},N}(F) = \frac{1}{\Phi} \sum_{\chi \in \mathfrak{A}} \chi(N+F) \overline{\chi(N)}. \quad (78)$$

Then

$$\boxed{\left\| \sum_{(N,F) \in \mathcal{I}} K_{\mathfrak{A},N}(F) z_{N,F} \right\|_{\mathcal{H}}^2 \leq \frac{|\mathfrak{A}|}{\Phi} \sum_r \|V_r\|_{\mathcal{H}}^2.} \quad (79)$$

If

$$\nu_{\text{rat}} = \max_r \#\{(N, F) \in \mathcal{I} : r(N, F) = r\}, \quad (80)$$

then

$$\left\| \sum_{(N,F) \in \mathcal{I}} K_{\mathfrak{A},N}(F) z_{N,F} \right\|_{\mathcal{H}}^2 \leq \frac{|\mathfrak{A}|}{\Phi} \nu_{\text{rat}} \sum_{(N,F) \in \mathcal{I}} \|z_{N,F}\|_{\mathcal{H}}^2. \quad (81)$$

Proof. Expand the left side of (77). Character orthogonality gives Φ when two pairs have the same ratio and zero otherwise. Regrouping the surviving inner products yields the bucket norms in (75). Next,

$$\sum_{N,F} K_{\mathfrak{A},N}(F) z_{N,F} = \frac{1}{\Phi} \sum_{\chi \in \mathfrak{A}} Y_\chi.$$

Cauchy in the character variable and (77) prove (79). Cauchy inside each ratio bucket gives $\|V_r\|^2 \leq \nu_{\text{rat}} \sum_{r(N,F)=r} \|z_{N,F}\|^2$, proving the last estimate. $\square$

For the full and proper families, the prefactors in (79) are respectively

$$\frac{|\mathfrak{X}_{\text{full}}|}{\Phi} = A = 1 - O(J^{-1}), \quad \frac{|\mathfrak{X}_{\text{prop}}|}{\Phi} = 1 - A \asymp J^{-1}. \quad (82)$$

The theorem gains the family density but leaves every coherent collision inside V_r untouched.

4.2 Exact fixed-determinant multiplicity

For fixed $F \bmod M$, let

$$\mathcal{D}_F = \{N \in (\mathbb{Z}/M\mathbb{Z})^\times : N + F \in (\mathbb{Z}/M\mathbb{Z})^\times\} \quad (83)$$

and define $r_F(N) = 1 + FN^{-1}$ on $\mathcal{D}_F$.

Theorem 4.2 (Exact target collision multiplicity). *The largest fiber of r_F has cardinality*

$$\boxed{d_M(F) = \prod_{q_i | F} (q_i - 1).} \quad (84)$$

Consequently, for arbitrary vectors $z_N \in \mathcal{H}$,

$$\sum_{\chi \bmod M} \left\| \sum_{N \in \mathcal{D}_F} z_N \chi(N+F) \overline{\chi(N)} \right\|_{\mathcal{H}}^2 \leq \Phi d_M(F) \sum_{N \in \mathcal{D}_F} \|z_N\|_{\mathcal{H}}^2. \quad (85)$$

The multiplicity and the inequality are sharp in the unrestricted Hilbert coefficient class.

Proof. Work first modulo q_i . If $F = 0 \pmod{q_i}$, then $r_F(N) = 1$ for all n_i unit values of N . If $F \neq 0 \pmod{q_i}$, the domain is $G_i \setminus \{-F\}$, and

$$N \longmapsto 1 + F/N$$

is a bijection onto $G_i \setminus \{1\}$, with inverse $N = F/(r - 1)$. Tensoring through the CRT gives (84). Apply [theorem 4.1](#) with fixed F to obtain (85). Equality in the bucket Cauchy step is obtained by taking equal vectors on a largest fiber. $\square$

If the primes satisfy $q_i \asymp J$, then

$$d_M(F) \asymp J^{|Z(F)|}, \quad Z(F) = \{i : q_i \mid F\}. \quad (86)$$

Thus the four determinant strata have collision scales

$$1, \quad J, \quad J^2, \quad J^3. \quad (87)$$

Generic F is the best possible case: the target ratio map is injective. The joint aliases $M \mid F$ are the worst possible case: every target is sent to $r = 1$.

4.3 Fixed-determinant face mass

The collision multiplicity may also be seen directly in the target mass of a face.

Proposition 4.3 (Exact target mass on a determinant stratum). *For $Z(F) = \{i : q_i \mid F\}$,*

$$\boxed{\sum_{N \bmod M}^* |C_N(F)|^2 = \prod_{i \in Z(F)} n_i(1 - p_i)^2 \prod_{i \notin Z(F)} (n_i - 1)p_i^2.} \quad (88)$$

If $M \nmid F$, then $P_N(F) = -C_N(F)$, so the same formula holds for the proper union. If $M \mid F$, then

$$\sum_{N \bmod M}^* |P_N(F)|^2 = \Phi(1 - A)^2. \quad (89)$$

For $q_i \asymp J$, the full-face scales in (88) are

$$J^{-3}, \quad J^{-1}, \quad J, \quad J^3 \quad (90)$$

as $|Z(F)| = 0, 1, 2, 3$, while the joint proper mass is $\asymp J$.

Proof. If $q_i \mid F$, the local full factor is $1 - p_i$ for all n_i unit targets. If $q_i \nmid F$, the target $N = -F$ is killed and the remaining $n_i - 1$ targets have local value $-p_i$. Tensor the local squared masses. The proper identities follow from (65) and (64). $\square$

Summing (88) over one F -period gives ΦA , and summing the proper mass gives $\Phi(1 - A)$, as required by fixed-target Parseval and Fubini. Over an integer determinant interval of length $O(X)$, these traces are at most $X^{1+o(1)}$ and $X^{1+o(1)}/J$, respectively. This trace bookkeeping does not control coherent row synthesis; the next section computes that synthesis exactly.

5 Exact target-synthesis spectrum

Fixed-target Parseval concerns the norm of one row. A target-coupled Bessel estimate concerns coherent combinations of many such rows. The distinction can be quantified without an inequality.

5.1 The local full-face Gram

Fix one odd prime q , put $n = q - 1$, $p = n^{-1}$, and let

$$c_N(F) = \mathbf{1}_{F=0} - p\mathbf{1}_{F \neq -N}, \quad N \in \mathbb{F}_q^\times, \quad F \in \mathbb{F}_q. \quad (91)$$

Lemma 5.1 (Exact local target Gram). *For $N, N' \in \mathbb{F}_q^\times$,*

$$\langle c_N, c_{N'} \rangle = \begin{cases} 1 - p, & N = N', \\ 1 - p - p^2, & N \neq N'. \end{cases} \quad (92)$$

Equivalently, the $n \times n$ Gram matrix is

$$G_q = p^2 I_n + (1 - p - p^2) \mathbf{1}\mathbf{1}^*. \quad (93)$$

Its squared singular values are

$$\lambda_q(\eta) = \begin{cases} \frac{q(q-2)^2}{(q-1)^2}, & \eta = \mathbf{1}, \\ \frac{1}{(q-1)^2}, & \eta \neq \mathbf{1}, \end{cases} \quad (94)$$

where η ranges over target Mellin modes.

Proof. For $N = N'$, the distinguished value $F = 0$ contributes $(1 - p)^2$, the killed value $F = -N$ contributes zero, and the remaining $q - 2 = n - 1$ values contribute p^2 . Their sum is $1 - p$.

For $N \neq N'$, the values $0, -N, -N'$ are distinct. The first contributes $(1 - p)^2$, the next two contribute zero, and the remaining $q - 3 = n - 2$ values contribute p^2 . The total is $1 - p - p^2$. This proves (93). On the orthogonal complement of $\mathbf{1}$, its eigenvalue is p^2 . On $\mathbf{1}$, the eigenvalue is

$$p^2 + n(1 - p - p^2) = \frac{q(q-2)^2}{(q-1)^2}.$$

□

The off-diagonal entry in (92) differs from the row norm by only p^2 . The local full rows are nearly parallel.

5.2 The CRT tensor spectrum

Return to three primes. Define the full target-synthesis operator

$$T_C : \ell^2((\mathbb{Z}/M\mathbb{Z})^\times) \longrightarrow \ell^2(\mathbb{Z}/M\mathbb{Z}), \quad (T_C a)(F) = \sum_{N \bmod M}^* a_N C_N(F). \quad (95)$$

For a target Mellin mode $\eta = \eta_1 \otimes \eta_2 \otimes \eta_3$, put

$$\lambda_C(\eta) = \prod_{i=1}^3 \lambda_{q_i}(\eta_i), \quad (96)$$

with the local factors in (94).

Theorem 5.2 (Complete full-face target spectrum). *The target Mellin modes diagonalize $T_C^* T_C$, and their squared singular values are exactly $\lambda_C(\eta)$. In particular,*

$$\|T_C\|_{\text{op}}^2 = \Lambda_C := \prod_{i=1}^3 \frac{q_i(q_i - 2)^2}{(q_i - 1)^2}. \quad (97)$$

If $q_i \asymp J$, then

$$\Lambda_C \asymp \Phi \asymp J^3. \quad (98)$$

Under the same hypothesis $q_i \asymp J$, every nonprincipal global target mode has squared singular value $O(1)$.

Proof. The CRT identifies T_C with the tensor product of the three local matrices in (91). Tensor lemma 5.1. The largest local eigenvalue is the principal one. If a global mode has a nonprincipal component at i , then that local factor is $O(J^{-2})$, while the two remaining principal factors are $O(J)$. Their product is $O(1)$. $\square$

The direct Gram formula is also useful. For $p_i = (q_i - 1)^{-1}$,

$$\langle C_N, C_{N'} \rangle = \prod_{i=1}^3 \left[(1 - p_i) \mathbf{1}_{N=N' \pmod{q_i}} + (1 - p_i - p_i^2) \mathbf{1}_{N \neq N' \pmod{q_i}} \right]. \quad (99)$$

If N and N' differ in every coordinate, the right side is $1 - O(J^{-1})$.

5.3 The proper target spectrum

Define

$$(T_P a)(F) = \sum_{N \bmod M}^* a_N P_N(F). \quad (100)$$

Since $P_N = \delta_0 - C_N$ and $C_N(0) = A$, the target Gram is

$$T_P^* T_P = T_C^* T_C + (1 - 2A) \mathbf{1} \mathbf{1}^*. \quad (101)$$

Theorem 5.3 (Complete proper-face target spectrum). *Every nonprincipal global target Mellin mode has squared singular value $\lambda_C(\eta)$. The global principal mode has squared singular value*

$$\lambda_P(\mathbf{1}) = \Lambda_C + \Phi(1 - 2A). \quad (102)$$

For $q_i \asymp J$,

$$\lambda_P(\mathbf{1}) \asymp \frac{\Phi}{J} \asymp J^2, \quad \|T_P\|_{\text{op}}^2 \asymp J^2. \quad (103)$$

Moreover,

$$\langle P_N, P_{N'} \rangle = 1 - 2A + \langle C_N, C_{N'} \rangle, \quad (104)$$

and

$$\langle C_N, P_{N'} \rangle = A - \langle C_N, C_{N'} \rangle. \quad (105)$$

Proof. Equation (101) follows by expanding $\langle \delta_0 - C_N, \delta_0 - C_{N'} \rangle$. The rank-one correction vanishes on every nonprincipal target mode and adds $\Phi(1 - 2A)$ on the principal mode. This proves the exact spectrum.

For the asymptotic, divide (102) by Φ . One has

$$\frac{\Lambda_C}{\Phi} = \prod_{i=1}^3 \left(1 - p_i - p_i^2 + p_i^3 \right), \quad (106)$$

while $A = \prod_i (1 - p_i)$. Expanding the fixed products shows

$$\frac{\lambda_P(\mathbf{1})}{\Phi} = \sum_{i=1}^3 p_i + O(J^{-2}) \asymp J^{-1}. \quad (107)$$

The row Gram identities follow from the same expansion. $\square$

5.4 Trace is not synthesis

The fixed-target row norms are

$$\|C_N\|_2^2 = A = 1 - O(J^{-1}), \quad \|P_N\|_2^2 = 1 - A \asymp J^{-1}. \quad (108)$$

By contrast, the synthesis norms are of orders J^3 and J^2 . Thus a diagonal or trace estimate over N cannot be upgraded to a target-coupled operator estimate without controlling the principal target direction. This is a precise finite-dimensional obstruction; it does not assert that the actual Möbius row vector occupies that direction.

Corollary 5.4 (Coefficient-blind target Bessel obstruction). *Any coefficient-blind inequalities*

$$\|T_C a\|_2^2 \leq B_C \|a\|_2^2, \quad \|T_P a\|_2^2 \leq B_P \|a\|_2^2 \quad (109)$$

valid for every scalar target vector a must satisfy

$$B_C \geq \Lambda_C \asymp J^3, \quad B_P \geq \lambda_P(\mathbf{1}) \asymp J^2. \quad (110)$$

Thus the fixed-row norms alone cannot furnish smaller synthesis constants.

Remark 5.5 (Scope). The theorem concerns unrestricted scalar target coefficients on one CRT period. Physical coefficients are far more structured. The result rules out a coefficient-blind frame argument; it is neither a lower bound for $V_L + V_R$ nor evidence that the actual Möbius vector is extremal.

6 Full/proper recombination and the joint-alias Gram

The preceding sections show why estimating the two face families separately is expensive. The CRT congruence-completed kernel contains their exact sum, so one should recombine before applying an inequality.

6.1 Strict face-splitting obstruction

Let $E \subset \mathbb{Z}/M\mathbb{Z}$ be a nonempty set of nonjoint residues on which $C_N(F) \neq 0$. Fix a unit vector v in a Hilbert space and set

$$z_F = \frac{\overline{C_N(F)}}{|C_N(F)|} v, \quad F \in E. \quad (111)$$

Proposition 6.1 (Separate faces are strictly stronger). *For the family (111),*

$$\sum_{F \in E} C_N(F) z_F = \left(\sum_{F \in E} |C_N(F)| \right) v, \quad (112)$$

and

$$\sum_{F \in E} P_N(F) z_F = - \left(\sum_{F \in E} |C_N(F)| \right) v. \quad (113)$$

Their recombined output is zero. Hence no abstract estimate can recover separate full and proper outputs from their sum.

Proof. Use $P_N(F) = -C_N(F)$ for $M \nmid F$. The definition of z_F aligns every full-face summand. $\square$

This proposition is not a counterexample to either fixed physical field estimate in (45): its vectors were chosen adversarially. It proves a logical point. Separate-face control is sufficient, but it is not necessary for the recombined physical kernel and may destroy an exact cancellation before arithmetic coefficients are used.

6.2 The identity bucket

Let $B \ll X$ bound the determinant range and put

$$\mathcal{I}_B = \{k \in \mathbb{Z} : |kM| \leq B\}, \quad K_B = |\mathcal{I}_B| = 2 \left\lfloor \frac{B}{M} \right\rfloor + 1. \quad (114)$$

Let

$$\mathcal{Z}_k := \mathcal{R}_{kM} \in \mathcal{H}_{\text{phys}}, \quad k \in \mathcal{I}_B, \quad (115)$$

be the raw shifted columns associated with

$$su = N + kM. \quad (116)$$

Define the two completed face outputs by

$$\mathcal{Y}_C := \sum_{|F| \leq B} \mathcal{Z}_F[C], \quad \mathcal{Y}_P := \sum_{|F| \leq B} \mathcal{Z}_F[P]. \quad (117)$$

Linearity and (63) give the exact identity

$$\boxed{\mathcal{Y}_C + \mathcal{Y}_P = \sum_{k \in \mathcal{I}_B} \mathcal{Z}_k =: \mathcal{Y}_{\text{alias}}.} \quad (118)$$

By contrast, the original equality output of the retained regular subpacket is

$$\boxed{\mathcal{Y}_{\text{phys}} = \mathcal{Z}_0 = \mathcal{R}_0.} \quad (119)$$

Thus face recombination produces an exact congruence completion of the original equality; it does not identify the completion with its zero column.

Proposition 6.2 (Character blindness on the identity bucket). *For every unit target N , every $k \in \mathcal{I}_B$, and every character $\chi \bmod M$,*

$$\chi(N + kM) \overline{\chi(N)} = 1. \quad (120)$$

Thus every joint alias belongs to the single ratio bucket $r = 1$, and the multiplicative character Gram on that bucket is rank one.

Proof. Modulo M , one has $N + kM = N$, and N is a unit. $\square$

No additional multiplicative character decomposition can separate the vectors in (118). The completion is governed by their ordinary Hilbert Gram matrix

$$\mathbf{G}_B = (\langle \mathcal{Z}_k, \mathcal{Z}_{k'} \rangle)_{k, k' \in \mathcal{I}_B}. \quad (121)$$

6.3 Alias completion does not recover exact equality

Theorem 6.3 (Sharp equality–recovery obstruction). *Assume $K_B > 1$. In the unrestricted Hilbert column class, neither $\mathcal{Y}_{\text{alias}}$ nor the pair $(\mathcal{Y}_C, \mathcal{Y}_P)$ controls $\mathcal{Z}_0$. More precisely, for every $T > 0$ there are columns for which*

$$\mathcal{Y}_C = \mathcal{Y}_P = \mathcal{Y}_{\text{alias}} = 0, \quad \|\mathcal{Z}_0\|_2 = T. \quad (122)$$

Proof. Choose $k_1 \in \mathcal{I}_B \setminus \{0\}$ and a unit vector v . Set

$$\mathcal{Z}_0 = Tv, \quad \mathcal{Z}_{k_1} = -Tv, \quad \mathcal{Z}_k = 0 \quad (k \neq 0, k_1), \quad (123)$$

and set every nonjoint target atom in every shifted output to zero. On a joint alias, $C_N(kM) = A$ and $P_N(kM) = 1 - A$, independently of N . Hence $\mathcal{Y}_C = A(\mathcal{Z}_0 + \mathcal{Z}_{k_1}) = 0$ and $\mathcal{Y}_P = (1 - A)(\mathcal{Z}_0 + \mathcal{Z}_{k_1}) = 0$, while $\|\mathcal{Z}_0\|_2 = T$. $\square$

The witness is coefficient–blind and artificial. It does not assert that the physical Möbius columns cancel in this way. It proves that an estimate for the two completed fixed fields needs a new physical recovery argument before it can bound (36).

6.4 Localized columns, the alias Gram, and the sharp Cauchy loss

Put

$$B_0 = Q^2 J. \quad (124)$$

The following strong localized input is useful for calibrating possible positive theorems.

Conditional input 6.4 (Uniform full alias–column bound). Uniformly for $k \in \mathcal{I}_B$,

$$\|\mathcal{Z}_k\|_2^2 \ll X^{o(1)} B_0. \quad (125)$$

At $k = 0$, [conditional input 6.4](#) already gives

$$\|\mathcal{Y}_{\text{phys}}\|_2^2 \ll X^{o(1)} Q^2 J, \quad (126)$$

which is stronger than the inherited target $X^{o(1)} Q^3 / J$, since $Q^2 J \mathcal{K}_X = X^{o(1)} Q^3 / J$. It is therefore not a preliminary consequence of the proved input–copy diagonal; it is itself a stronger coefficient theorem for the raw output.

Conditional input 6.5 (Joint–alias Gram almost–orthogonality). The Gram matrix in (121) satisfies

$$\boxed{\|\mathbf{G}_B\|_{\text{op}} \ll X^{o(1)} B_0.} \quad (127)$$

Proposition 6.6 (What the strong Gram gate implies). Under [conditional input 6.5](#),

$$\left\| \sum_{k \in \mathcal{I}_B} \mathcal{Z}_k \right\|_2^2 \ll X^{o(1)} B_0 K_B. \quad (128)$$

At the physical endpoint $B \asymp X$, the right side has scale

$$X^{o(1)} B_0 K_B \asymp X^{o(1)} Q^2 J \mathcal{K}_X = X^{o(1)} \frac{Q^3}{J}. \quad (129)$$

Moreover,

$$\|\mathcal{Z}_k\|_2^2 \leq \|\mathbf{G}_B\|_{\text{op}} \ll X^{o(1)} B_0 \quad (k \in \mathcal{I}_B). \quad (130)$$

In particular, the Gram gate implies [conditional input 6.4](#) and the stronger equality bound (126).

Proof. The left side of (128) is $\langle \mathbf{G}_B \mathbf{1}, \mathbf{1} \rangle$. Bound it by $\|\mathbf{G}_B\|_{\text{op}} \|\mathbf{1}\|_2^2$. The diagonal entries of a positive semidefinite Gram matrix are bounded by its operator norm, proving (130). $\square$

For the alias-completed vector alone, the operator-norm gate is stronger than necessary. Given the diagonal trace bound, a weaker sufficient one-vector condition is

$$\operatorname{Re} \sum_{k \neq k'} \langle \mathcal{Z}_k, \mathcal{Z}_{k'} \rangle \ll X^{o(1)} B_0 K_B, \quad (131)$$

together with the diagonal trace of the same size. We retain (127) because it is the genuine Bessel form.

Proposition 6.7 (Sharp coefficient-free alias loss). *Suppose only the diagonal trace estimate*

$$\sum_{k \in \mathcal{I}_B} \|\mathcal{Z}_k\|_2^2 \ll X^{o(1)} B_0 K_B. \quad (132)$$

Then Cauchy gives

$$\left\| \sum_{k \in \mathcal{I}_B} \mathcal{Z}_k \right\|_2^2 \ll X^{o(1)} B_0 K_B^2. \quad (133)$$

The factor K_B relative to (128) is optimal for the abstract Hilbert column class.

Proof. Cauchy proves (133). For sharpness, let v be a unit vector and take

$$\mathcal{Z}_k = \sqrt{B_0} v \quad (k \in \mathcal{I}_B). \quad (134)$$

Then the diagonal trace is $B_0 K_B$, while the squared sum is $B_0 K_B^2$. $\square$

At $B \asymp X$, the exact finite loss in proposition 6.7 is K_B , and

$$\boxed{K_B \asymp \mathcal{K}_X := 1 + \frac{X}{M} = \frac{X}{J^3} X^{o(1)} = \frac{Q}{J^2} X^{o(1)} = X^{1/400+o(1)}}. \quad (135)$$

This is much smaller than the earlier J -scale face losses, but it is a fixed power and cannot be hidden in $X^{o(1)}$.

Remark 6.8 (The old diagonal is a different object). Equation (37) proves the input-copy diagonal, not $\|\mathcal{R}_0\|_2^2$. For $k \neq 0$, the affine shift is $h + kM$, which may be of size X , and the fixed- h primitivity and incidence hypotheses are not automatically uniform. Thus conditional input 6.4 is both arithmetically new and, at $k = 0$, already stronger than the inherited physical target.

6.5 A target-scale recovery formulation

The original gate can be recovered without a full Gram theorem if one can control two particular vectors rather than every alias coefficient.

Proposition 6.9 (Two-sum equality recovery). *Suppose*

$$\|\mathcal{Y}_{\text{alias}}\|_2^2 \ll X^{o(1)} B_0 K_B, \quad (136)$$

$$\left\| \sum_{\substack{k \in \mathcal{I}_B \\ k \neq 0}} \mathcal{Z}_k \right\|_2^2 \ll X^{o(1)} B_0 K_B. \quad (137)$$

Then

$$\|\mathcal{Z}_0\|_2^2 \ll X^{o(1)} B_0 K_B. \quad (138)$$

At the endpoint, this is the inherited scale $X^{o(1)} Q^3 / J$.

Proof. Subtract the nonzero–alias sum from $\mathcal{Y}_{\text{alias}}$ and use $\|x - y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$. $\square$

If the two fixed–field estimates (45) were proved, their period ledgers and (118) would imply (136). They do not imply (137), because the fixed fields are not assumed to be stable under deleting $F = 0$. This is the precise additional recovery gate missing from the earlier sufficient–input claim.

6.6 A fourth modulus removes aliases but not the equality gate

Choose four pairwise coprime primes $q_i \asymp J$ and put $M_4 = \prod_{i=1}^4 q_i$. Then $M_4 > X$ at the endpoint, so

$$\mathbf{1}_{M_4|F} = \mathbf{1}_{F=0} \quad (|F| \ll X). \quad (139)$$

This removes every nonzero joint alias. However, on $F = 0$ the four–coordinate full face has scalar value

$$A_4 = \prod_{i=1}^4 \left(1 - \frac{1}{q_i - 1}\right) = 1 - O(J^{-1}). \quad (140)$$

Corollary 6.10 (Four–modulus loopback). *Adding a fourth orbit–scale modulus eliminates the alias Gram but returns the full face, up to the scalar A_4 , to the original equality operator. It supplies no coefficient–free power saving for the full physical gate.*

This is not a negative result about the physical coefficient. It shows that the alias problem and the equality coefficient problem are distinct: more CRT coordinates solve the former and leave the latter unchanged.

7 Terminal embeddings and sparse Möbius progressions

The identity bucket has an explicit arithmetic submodel. We record it to calibrate the strength of possible operator formulations, not to assert a new Möbius estimate.

7.1 Terminal aliases

On the terminal factorization $s = 1$, the k -th joint alias has

$$u = N + kM. \quad (141)$$

For a smooth terminal cutoff Φ_0 , define

$$a_N(k) = -\mu(N + kM) \log(N + kM) \Phi_0(N + kM). \quad (142)$$

For every finite interval I of admissible k 's,

$$\begin{aligned} \left| \sum_{k \in I} a_N(k) \right|^2 &= \sum_{k \in I} |a_N(k)|^2 \\ &\quad + 2 \operatorname{Re} \sum_{r \geq 1} \sum_{k, k+r \in I} a_N(k+r) \overline{a_N(k)}. \end{aligned} \quad (143)$$

The off–diagonal terms contain

$$\mu(N + kM) \mu(N + (k+r)M) \quad (144)$$

at shifts rM . The progression has modulus $M = X^{399/400+o(1)}$ and length only $\mathcal{K}_X = X^{1/400+o(1)}$.

Proposition 7.1 (Exact arithmetic content of the terminal alias Gram). *The same-target part of the joint-alias Gram contains the autocorrelation (143). Restoring the physical row $\alpha = (\ell, d)$ and target $N_\alpha(j) = \ell d j + h$, the cross-row terms contain weights of the form*

$$\begin{aligned} & \mu(d_\alpha) \mu(d_\beta) (\log \ell_\alpha) (\log \ell_\beta) \\ & \times \mu(N_\alpha(j) + kM) \mu(N_\beta(j) + k'M), \end{aligned} \quad (145)$$

multiplied by the actual masks and smooth factors.

Proof. Insert (142) into the Gram expansion. The physical row coefficient contains $\mu(d) \log \ell$; pairing two rows gives (145). $\square$

Equation (145) is the shifted version of the affine four-Möbius terminal boundary in TPC-34. The required physical theorem would average it over rows, orbit times, opposite rows, aliases, and projective layers. It is therefore not logically equivalent to proving $\sum_{k < K} \mu(N + kM) = o(K)$ for every N . Conversely, it cannot be deduced from character orthogonality because all joint aliases have ratio one.

7.2 Prescribed fields versus a hereditary all-field estimate

Let $\mathcal{K}$ be a coefficient Hilbert space containing the two prescribed fields C, P , and let $\mathfrak{T} : \mathcal{K} \rightarrow \mathcal{H}_{\text{phys}}$ be the linear synthesis map

$$\kappa \mapsto \sum_F \mathcal{Z}_F[\kappa]. \quad (146)$$

The TPC-37 continuation asks only for bounds at the two fields C and P . An all-field estimate of the same normalization would assert

$$\left\| \sum_F \mathcal{Z}_F[\kappa] \right\|_2^2 \ll X^{o(1)} B_0 \sup_N \sum_F |\kappa_N(F)|^2 \quad (147)$$

for every admissible field κ . We call it hereditary if the same bound remains valid after multiplying the field by indicators of determinant fibers or factorization layers. This norm is the one appearing in the earlier prescribed-field input; no Hilbert structure for the right-hand side is being asserted.

Proposition 7.2 (Two tests do not determine an operator norm). *Bounds for $\mathfrak{T}C$ and $\mathfrak{T}P$ alone give no bound for $\|\mathfrak{T}\|_{\text{op}}$ on the orthogonal complement of their span in $\mathcal{K}$. Moreover, a small recombined value $\mathfrak{T}(C + P)$ gives no bound for either separate value.*

Proof. The first statement is elementary finite-dimensional linear algebra: an operator may vanish on two prescribed vectors and have arbitrary norm on their orthogonal complement. For the second, take $\mathfrak{T}C = v$ and $\mathfrak{T}P = -v$, or use the explicit offjoint construction in proposition 6.1. $\square$

In addition, theorem 6.3 shows that the two completed test values do not recover their $F = 0$ column. This is logically distinct from their failure to determine the operator norm. The distinction becomes arithmetic on the equality face. Heredity under $F = 0$ would in particular give the explicit restricted input

$$\|\mathcal{Z}_0[\kappa]\|_2^2 \ll X^{o(1)} B_0 \sup_N |\kappa_N(0)|^2 \quad (148)$$

for every field supported on $F = 0$.

Theorem 7.3 (Equality–face embedding). *Assume (148), and let $\mathcal{R}_0 := \mathcal{Z}_0[1]$ be the raw regular–subpacket output on $F = 0$. Then*

$$\boxed{\mathcal{Z}_0[C] = A\mathcal{R}_0, \quad \mathcal{Z}_0[P] = (1 - A)\mathcal{R}_0.} \quad (149)$$

Consequently, the hereditary all–field estimate at normalization $B_0 = Q^2J$ implies

$$\|\mathcal{R}_0\|_2^2 \ll X^{o(1)}Q^2J. \quad (150)$$

This is stronger than the inherited target Q^3/J by a factor of order $\mathcal{K}_X = (Q/J^2)X^{o(1)}$.

Proof. The identities follow from $C_N(0) = A$ and $P_N(0) = 1 - A$, which are independent of N . Apply the hereditary estimate to $C_N(F)\mathbf{1}_{F=0}$ and divide by $A^2 = 1 - O(J^{-1})$. Finally, $Q^2J\mathcal{K}_X = X^{o(1)}Q^3/J$ by (48). $\square$

If the hereditary all–field estimate is also stable under the terminal restriction $s = 1$, then (150) applies to the affine terminal operator whose square expands into (145) with $k = k' = 0$.

Corollary 7.4 (Terminal hereditary all–field implication). *An all–field estimate hereditary under $F = 0, s = 1$ implies a diagonal–scale bound for the terminal affine–Möbius operator isolated in TPC–34.*

This corollary does not say that the two prescribed–field inputs (45) implies affine Chowla. By itself that input does not even recover the equality column from its alias completion, and it may also use cancellation between equality, proper divisors, terminal atoms, and nonzero aliases. The implication applies only to the stronger, localizable operator theorem.

7.3 What the sharp examples do not prove

The rank–one witness (134), the offjoint witness (111), and the principal target mode in section 5 are artificial coefficient choices. They establish optimal losses in broad abstract classes. They do not prove that the actual Möbius vectors align, that conditional input 6.5 is false, or that a Möbius autocorrelation is large. Their role is to identify precisely which new arithmetic information a positive theorem must use.

8 The corrected route tree: completion and equality recovery

The target–coupled continuation separates into exact CRT algebra, a congruence–completed analytic output, and an equality–recovery step. The last step cannot be omitted. This section records the corrected route tree.

8.1 What the exact algebra removes

The unconditional reductions prove the following.

- (i) The dependence on a moving target N is exactly the ratio phase $\chi(N + F)\overline{\chi(N)}$; no unidentified affine shift remains in the finite CRT algebra.
- (ii) Fixed–target character orthogonality, moving–target shifted Jacobi Grams, and every zero–extended character convention are explicit.
- (iii) For fixed F , all coefficient–free target collisions are localized by the exact multiplicity $d_M(F)$. Generic determinant strata have no target collision.

- (iv) Full/proper face splitting may be reversed before estimating. Every nonjoint ghost then cancels identically, and the completed output is the joint–alias sum $\mathcal{Y}_{\text{alias}}$.
- (v) The completion has $O(\mathcal{K}_X)$ columns $F = kM$, with $\mathcal{K}_X = X^{1/400+o(1)}$ at the endpoint. On them, every multiplicative character phase is one.
- (vi) Diagonal trace plus coefficient–blind Cauchy loses exactly the finite alias count K_B on the completed alias sum; at the endpoint this has order $\mathcal{K}_X$, and the loss is sharp in the abstract Hilbert column class.
- (vii) The original physical output is the single column $\mathcal{Z}_0$, not $\mathcal{Y}_{\text{alias}}$. The completed fixed fields do not recover this column in the abstract class.

The corrected implication ledger is therefore

$$\left. \begin{array}{l} \text{two fixed face–field estimates} \\ \text{plus the exact period ledgers} \end{array} \right\} \Rightarrow \|\mathcal{Y}_{\text{alias}}\|_2^2 \ll X^{o(1)} Q^2 J \mathcal{K}_X, \quad \mathcal{Y}_{\text{alias}} \not\approx \mathcal{Z}_0. \quad (151)$$

The final nonimplication is witnessed by [theorem 6.3](#).

8.2 Route A: target–scale two–sum recovery

The weakest explicit completion–based recovery route isolated here is to prove the completed bound in [\(136\)](#) and, independently, the nonzero–alias aggregate bound

$$\left\| \sum_{\substack{k \in \mathcal{I}_B \\ k \neq 0}} \mathcal{Z}_k \right\|_2^2 \ll X^{o(1)} Q^2 J K_B. \quad (152)$$

Their difference is $\mathcal{Z}_0$, so [proposition 6.9](#) gives the inherited target $X^{o(1)} Q^3 / J$.

This route does not ask for uniform control of every alias column or for an operator estimate under arbitrary alias weights. It asks only for one additional physical vector. Its expansion contains the averaged M -spaced Möbius correlations in [proposition 7.1](#); hence it still requires actual coefficient information, but it is weaker than pointwise Chowla and weaker than a hereditary all–field theorem.

8.3 Route B: strong localized or Gram theorems

One may instead prove the uniform raw–column estimate

$$\|\mathcal{Z}_k\|_2^2 \ll X^{o(1)} Q^2 J \quad (k \in \mathcal{I}_B). \quad (153)$$

This is a meaningful moving–shift theorem, because the shifts

$$h_k = h + kM \quad (154)$$

have new primitivity and incidence behavior. It is not merely a uniform version of the old input–copy diagonal: at $k = 0$, [\(153\)](#) already proves the original physical output at the stronger $Q^2 J$ scale.

The still stronger operator route is

$$\|\mathbf{G}_B\|_{\text{op}} \ll X^{o(1)} Q^2 J. \quad (155)$$

Since a Gram operator controls its diagonal entries, [\(155\)](#) implies [\(153\)](#). These are therefore nested alternatives, not two independent sequential gates. The Gram theorem is reusable under arbitrary alias weights; the localized theorem is already sufficient for the original output.

8.4 Route C: exact recovery by several completions

A different possibility is to construct several CRT completions with moduli M_d and weights w_d such that, throughout the determinant window,

$$\sum_d w_d \mathbf{1}_{M_d|F} = \mathbf{1}_{F=0}. \quad (156)$$

If each weighted completion can be controlled without an excessive synthesis norm, this identity recovers equality algebraically rather than by subtracting a nonzero–alias sum. No such well–conditioned family is constructed here.

The four–modulus case is the simplest exact recovery: $M_4 > X$ makes $\mathbf{1}_{M_4|F} = \mathbf{1}_{F=0}$. However, its full face has coefficient $1 - O(J^{-1})$ at equality, so the method loops back to the unsuppressed original coefficient. More moduli solve recovery but do not by themselves solve the arithmetic energy estimate.

8.5 Three viable outcomes for the next layer

Positive aggregate recovery theorem.

Prove (152) for the actual rows and combine it with a completed fixed–field estimate. This is the most economical route identified by the present audit.

Positive strong theorem.

Prove the localized column estimate or the alias–Gram estimate. Either would be independently publishable; the latter is strictly stronger and includes the former.

Scoped obstruction or recovery no–go.

Within a precisely stated physical coefficient, shift, or completion class, prove that no target–scale recovery functional exists, or that a principal alias direction is unavoidable. Such a result would rule out that architecture at the stated normalization, not prove a lower bound for the prime–pair remainder.

8.6 Relation to the parity boundary

The exact rank–one directions and terminal Möbius correlations are consistent with the classical parity obstruction for sieve methods [2, 3]. This paper neither reformulates nor resolves that obstruction. A future positive recovery theorem would have to exploit coefficient information beyond the finite character identities proved here. Even then, it would first establish the inherited energy gate; further work would still be required before any prime–pair asymptotic.

9 Exact certificate and formal claim boundary

The accompanying standard–library program is

```
experiments/tpc38_certificate.py.
```

It writes the canonical archive

```
experiments/tpc38_certificate.json.
```

The same suite is run with ordinary Python and with optimization enabled:

```
python experiments/tpc38_certificate.py
python -O experiments/tpc38_certificate.py
```

No check depends on `assert` semantics.

9.1 Finite identities checked

The certificate uses integers and rational numbers only. It checks:

- (i) the local and CRT full/proper face definitions and exact recombination $C_N + P_N = \mathbf{1}_{M|F}$;
- (ii) fixed-target one-period squared norms;
- (iii) the local and CRT target-row Gram formulas;
- (iv) the local target-synthesis eigenvalues and the global tensor product ledger;
- (v) the exact ratio-fiber multiplicity $d_M(F)$ on every determinant stratum for several small prime packets;
- (vi) the fixed-determinant full/proper target masses;
- (vii) the four-modulus no-wrap threshold on a finite interval;
- (viii) offjoint full/proper cancellation; and
- (ix) the sharp K_B -loss rank-one alias witness and the two-column equality-nonrecoverability witness.

Finite enumeration verifies the displayed algebra on the archived test families. It is not evidence for an asymptotic Möbius or Gram estimate.

9.2 Reproducibility record

The final JSON records per-family check counts, byte-for-byte SHA-256 hashes of the tracked sources, and a digest of the canonical certificate payload. For the archived run:

Certificate source SHA-256.

1f2f479b8ff1891eebd379f1357d9fc03009dfe2583b68a982ea20d28409dc42

JSON SHA-256.

9750d268eec35885d034ce12b0be52c83e9008f9a8a515d49ad0a72bc967c3d6

Certificate digest.

f7a09acd95d4e5ff4420da353c04daded1c679b1c43da6e1b3bc6492da0c8f2a

These values were inserted after the hashed manuscript sources and certificate were frozen. The certificate hashes do not promote finite enumeration to a proof of the analytic reductions.

9.3 Unconditional conclusions

Under the hypotheses stated in the corresponding results (and using the inherited notation only when physical scales are mentioned), this paper proves:

- (i) the exact target-coupled character expansion of every CRT face for unit targets;
- (ii) full/proper recombination and the elimination of all nonjoint ghosts in the regular CRT completion before estimation;
- (iii) fixed-target Parseval and the shifted Jacobi cross-target Gram;
- (iv) Hilbert-valued ratio Parseval and the exact determinant multiplicity $d_M(F)$, hence the scale ladder $1, J, J^2, J^3$ when $q_i \asymp J$;
- (v) exact fixed-determinant target masses;
- (vi) the complete scalar target-synthesis spectra of the full and proper fields;
- (vii) character blindness on the joint-alias identity bucket;

- (viii) the sharp coefficient-free K_B Cauchy loss for coherent alias columns;
- (ix) nonrecoverability of the exact equality column from the completed alias sum or the two completed fixed face fields;
- (x) strict non-equivalence between separate face estimates and their recombined output;
- (xi) the equality-face embedding and the strength of a hereditary all-field operator estimate; and
- (xii) the exact terminal alias autocorrelation identity and the displayed four-Möbius factor contained in the physical row kernel.

9.4 What is not proved

Neither the manuscript nor the certificate proves:

- (i) the uniform full alias-column bound (125);
- (ii) the joint-alias Gram estimate (127) or its one-vector weakening;
- (iii) the two fixed physical face-field estimates (45);
- (iv) the nonzero-alias recovery estimate (137), or any other recovery of $\mathcal{Z}_0$ from the completed face outputs;
- (v) a hereditary target-coupled all-field operator theorem;
- (vi) the full physical orbit gate $V_L + V_R \ll X^{o(1)}Q^3/J$;
- (vii) cancellation in the Möbius progression $N + kM$, pointwise or on average;
- (viii) an affine Chowla theorem or a new Möbius-disjointness theorem;
- (ix) failure of sieve parity or positivity;
- (x) a Hardy-Littlewood prime-pair asymptotic; or
- (xi) infinitely many twin primes.

The advance is an exact reduction and a corrected logical gate. The finite unit-target CRT dependence and regular nonjoint face geometry have been resolved; the remaining physical identity bucket has a target-scale recovery route, a nested stronger alias-Gram route, and a sharp coefficient-free loss.

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